

✦ ASTRONOMY & ASTROPHYSICS

Mapping signals from the radio sky.

I'm Niriantsoa — an M.Sc. student working at the intersection of astronomy, data and code. Based in Antananarivo, Madagascar, I spend my time on observational astronomy, scientific computing, and turning raw signals into something legible.

See my work

Get in touch



ABOUT

I'm a Master's student in Astronomy and Astrophysics at the University of Antananarivo, currently training in radio astronomy and VLBI data reduction through the DARA programme. My work sits between observational astronomy and scientific computing — I like turning messy datasets into something clear. Outside coursework, I build small Python tools and explore how machine learning could help with radio astronomy data, including MeerKAT.

BASED IN
Antananarivo, Madagascar

FOCUS
Radio astronomy & scientific computing

CURRENTLY
DARA Basic Training, 2025–2026

SELECTED WORK

PythonAstropyPySide6

Moon Phase Calculator

A desktop application using Astropy to calculate lunar phases and illumination, with a PySide6 interface.

View on GitHub →

PythonPandasMatplotlib

IMDb Movie Dataset Analysis

An ETL and exploratory data analysis pipeline over a large IMDb movie dataset, answering key analytical questions with Pandas and Matplotlib.

View on GitHub →

TOOLKIT

PROGRAMMING

PythonNumPyPandas
MatplotlibAstropyLinux
Git/GitHub

ASTRONOMY

CASASAOImage DS9
Astrometrica
Radio InterferometryVLBI
FITS Data Analysis

DATA ANALYSIS

Scientific Computing
Data Visualization
Data Pre-processing

PATH

Ongoing
Master 2 (M.Sc.) in Astronomy & Astrophysics
University of Antananarivo — coursework in Galaxies, Cosmology, Python Programming, Linux

2025–2026
DARA Basic Training Programme
Development in Africa with Radio Astronomy — Linux & Python for scientific computing, optical & radio astronomy, VLBI data reduction

2021–2024
B.Sc. in Physics & Chemistry Education
École Normale Supérieure (ENS), Antananarivo — graduated with honours (14/20)

CURRENTLY EXPLORING

Radio AstronomyMachine Learning
Artificial IntelligenceScientific Computing
Signal ProcessingAstronomical Data Analysis

TALKS & TRAVEL

DARA Annual Network MeetingSouth Africa, 2026
TART ConferenceMadagascar, 2026

LANGUAGES

Malagasy — NativeFrench — Fluent
English — Professional

SAY HELLO

Always glad to talk radio astronomy, data — or anything in between.

The fastest way to reach me is by email.

niriantsoa1304@gmail.com

GitHub